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e.g.:- Express the statement "Everyone has exactly one best friend" as a logical expression involving predicates, quantifiers with a domain consisting of all people, and logical connectives.

Ans.:- $\forall x \exists y (B(x, y) \wedge \forall z ((z \neq y) \rightarrow \neg B(x, z)))$

e.g.:- Use quantifiers to express the statement "There is a woman who has taken a flight on every airline in the world."

Ans.:- $\exists w \forall a \exists f (P(w, f) \wedge Q(f, a)).$

OR

$\exists w \forall a \exists f R(w, f, a)$

Where $P(w, f)$: w has taken f .

$Q(f, a)$: f is a flight on a .

$R(w, f, a)$: w has taken f on a .

Also the domain of discourse for w , f and a consists of all the women in the world, all airplane flights and all airlines, respectively.

Note:-

$$P \rightarrow Q$$

$$\therefore Q \rightarrow P$$

[Converse error]

$$P \rightarrow Q$$

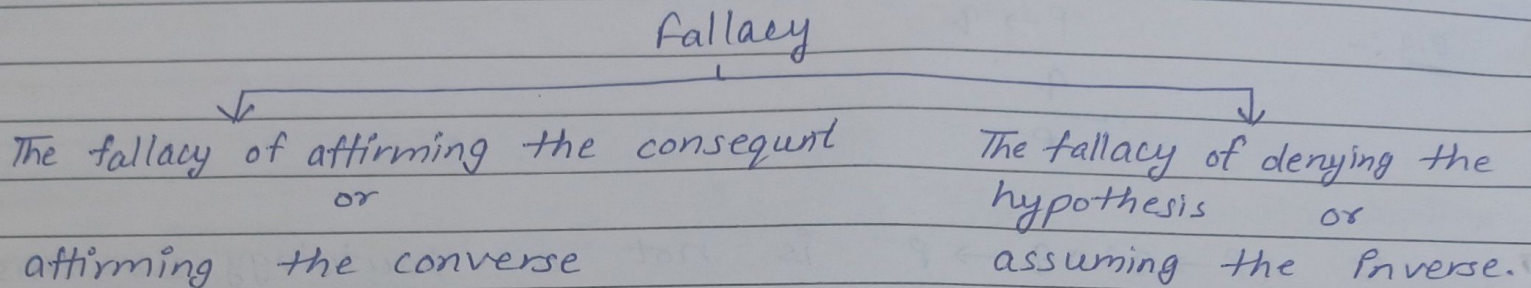
$$\therefore \neg P \rightarrow \neg Q$$

[Inverse error]

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Fallacies :-

A fallacy is an error in reasoning that results in an invalid argument.



e.g:-

$$P \rightarrow Q$$

$$Q$$

$$\hline P$$

[Modus Ponens is not applied here]

If Ravi solved this problem, then he obtained the answer 5.

Ravi obtained answer 5.

Therefore, Ravi solved this problem correctly.

Above argument is of the form: $P \rightarrow Q$ and Q then P .

This argument is invalid because the conclusion can be false even though $P \rightarrow Q$ and Q are true.

$((P \rightarrow Q) \wedge Q) \rightarrow P$ is not a tautology.

This is termed as the fallacy of affirming the conclusion.

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e.g.:- Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Sol.:- Let P : You did every problem in this book.

Q : You learned discrete mathematics.

According to the question,

$$\begin{array}{r} P \rightarrow Q \\ \neg Q \\ \hline \neg P \end{array}$$

The proposition $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$ is not a tautology, because it is false when P is false and Q is true. This is termed as the fallacy of denying the hypothesis.

Methods of proving theorems

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Direct Proofs

A direct proof of a conditional statement $P \rightarrow Q$ is constructed when the first step is the assumption that P is true, subsequent steps are constructed using rules of inference with the final step showing that Q must also be true.

Def. :- The integer n is even if there exists an integer k such that $n = 2k$.

And n is odd if there exists an integer k such that $n = 2k + 1$.

Note :- An integer can either be even or odd but not both.

Ques :- Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd".

Sol. :- The theorem states that $\forall n (P(n) \rightarrow Q(n))$
where $P(n)$: n is an odd integer.
 $Q(n)$: n^2 is an odd integer.

By the definition of odd integer,
 $n = 2k + 1$, where k is some integer.

Squaring on both sides

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

$$n^2 = 2(2k^2 + 2k) + 1$$

Let $2k^2 + 2k = t$, where t is an integer.

$$n^2 = 2t + 1$$

so, n^2 is also an odd integer.

e.g:- Give a direct proof that "if m and n are both perfect squares then mn is also a perfect square"

Solⁿ:- We assume m and n both are perfect squares. By the def. of perfect square it follows that there are integers s and t such that

$$m = s^2 \quad \text{and} \quad n = t^2$$

$$\begin{aligned} \text{then, } mn &= s^2 t^2 \\ &= (st)^2 \quad [\text{Using Commutivity and associativity of multiplication}] \end{aligned}$$

$\therefore mn$ is also a perfect square.